

# Ecosystemas: Representing Ecosystem Impacts in Design

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## ABSTRACT

*Personas*—distilled representations of particular user groups—are a key part of many design processes. Researchers have created *animal personas* to represent nonhuman animals as stakeholders in design efforts. However neither human nor animal personas allow for the representation of broader-scale ecological impacts in design decisions. To address this gap, we present an additional conceptual tool for design: the *ecosystema*. This design construct is analogous to a persona, but at the level of an entire ecosystem rather than of a particular human population or animal species. These constructs could allow ecosystem-wide impacts to influence design processes more effectively, and may provide greater leverage on current environmental crises than existing human-centered techniques.

## CCS CONCEPTS

• **Social and professional topics** → **Sustainability**; • **Human-centered computing** → **Human computer interaction (HCI)**.

## KEYWORDS

Sustainability, design, human-computer interaction, ecosystem, persona.

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## 1 INTRODUCTION

Across the past several centuries, human population has grown dramatically, from less than 1 billion in 1800 to over 7.7 billion people at present [45]. Human energy use during the same period has increased as well, more than tripling per capita and growing 25-fold in total [85]. Human exploitation of Earth's ecosystems—extracting

resources and co-opting ecosystem services for humanity's own use—has made these increases possible.

During the same period, Earth's biodiversity has decreased dramatically, with extinctions now occurring at up to 100 times the typical rate across the last 2 million years [17]. Researchers have estimated that it will take 3 million to 7 million years for mammals to evolve new species to replace those humans have driven to extinction [104]. The suffering of organisms of nonhuman species has increased dramatically as well, across both wild and domesticated species [15]. As humans co-opt ecosystems for their own use—converting forests to farmland, marshes to malls—species that had previously benefited from those ecosystems are no longer able to do so, typically dying or migrating as a result.

We argue that a serious limitation of much current design knowledge arises from its embedding in an anthropocentric framing [98]. Approaches such as “user-centered design” [96], “human-centered computing” [73], and “relationship-centered design” [23] typically place people at the center of the design process. These approaches arise from a shared anthropocentric model of the world that puts human needs and wants above the needs and wants of other species. However, a non-anthropocentric framing could produce a quite different body of design knowledge. We propose that a non-anthropogenic framing could help create design knowledge with greater leverage on the crises currently facing the Earth [14, 46] than current human-centered techniques.

In this paper, we present a new conceptual design tool, the *ecosystema* (see Figure 1 for a prototype). This construct is analogous to a persona [69, 79] but at the level of an entire ecosystem. By presenting this new conceptual tool to support design processes, we seek to enable human designs to contribute to the reversal of the harm that humans have done to the ecosystems of which we are a part.

## 2 RELATED WORK

The research described here brings together a range of domains, integrating work in social ecology with research in computing and design.

Various scholars in social ecology have explored the relationship between humans and nonhumans. For example, Helmut Haberl and his colleagues examined relationships between society and nature over time [40]. Social ecologist Dan Stokols' paper on directions of environmental design research in the Anthropocene [89] argued

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# Amazon Rainforest



## Endangered Species

|                                                                                    |                                    |
|------------------------------------------------------------------------------------|------------------------------------|
|   | 40,000<br>plant species            |
|   | 16,000<br>tree species             |
|   | 3,000<br>freshwater fish species   |
|   | 1,300<br>bird species              |
|  | more than 370<br>types of reptiles |

## Conservation

### Science and Technology:

- Real-time deforestation monitoring - drones to detect and report threats
- Conservation Hubs - biodiversity monitoring with camera traps, and environmental DNA analysis
- Advancing Science and Education through research of ecological health and climate change

### Safeguard Territories:

- Strengthen management of existing protected areas
- Further economic benefits and natural capital accounting to avoid further deforestation
- Promote sustainable agriculture to increase production of forest products and protect from illegal logging and gold mining threats

## USER STORY

This brilliant 1.4 billion-acre forest, spanning 9 countries, is among the most important terrestrial biomes. The forest offers a homeland, resources, and protection to 400 indigenous tribes. The U.S. National Cancer Institute has noted that 70% of the plants useful in cancer treatments grow in rainforest. Additionally, a great deal of biodiversity is found here, including 10% of the world's species. One hundred new species are discovered yearly in the Amazon. This rainforest absorbs at least 5% of global annual carbon emissions. 70% of South America's GDP is supported by the water found here, as well as \$8.2 billion towards Brazil's economy.

## Deforestation

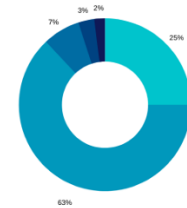


At this current rate of deforestation, the WWF expects 27% of biome tree-loss by 2030. There is a large correlation between the growth of South American infrastructure, such as building roads and increasing energy efforts, and the loss of forest area. Cattle ranching and development in soy plantations exported as beef and soy accounts for 1/5 of the destruction as land is opened for grazing and plantations.

## Climate Change

Rising temperatures have led to reduced rainfall across the rainforest. The lack of rain causes drought and increases the susceptibility of the Amazon to fire. If climate change continues to be neglected, it will be no surprise if the Amazon rainforest transitions into the Amazon savanna.

## Causes of Forest Loss



|                         |     |
|-------------------------|-----|
| cattle ranching         | 63% |
| small scale agriculture | 25% |
| commercial agriculture  | 7%  |
| logging                 | 3%  |
| other                   | 2%  |

Figure 1: An ecosystema for the Amazon rainforest.

for a broader conceptualization of environmental “users” - including, for example, residents geographically removed from the direct impacts of a particular building or other designed structure, whose health may be impacted by the carbon emissions or air pollution emitted at the site of origin due to atmospheric or marine transport of carbon and pollutants emitted at one site to more distant “telecoupled” regions [88]. (See also [58, 59] for further articulations of telecoupling and systems integration.) The “deep ecology” movement embraces the principle of “biocentrism” or “biocentric egalitarianism” (vs. anthropocentrism) and is premised on the idea that humans must be decentered, or relegated to a less powerful and preeminent role in ecosystems, if those systems and the diverse species that comprise them are to survive and thrive (see for example [24, 61, 64]). As Stokols has written: “[a]nthropocentrism prioritizes human needs over those of other species...Deep ecologists believe that biospheric harmony can be regained only if people reject their anthropocentric biases and embrace *biocentric egalitarianism*. *Biocentrism* holds that all life forms have an ‘equal right to live and blossom’ ([64], p.96).” ([88], p.267) Social ecologist Murray Bookchin has written about the evolution from biological and societal nature to “thinking nature” in which humans’ reasoning capacities would be applied to ecosystem design in ways that promote more equitable relationships between humans and other species [10, 11]. The ecosystema concept builds on these ideas and

broadens this perspective to include the interests and rights of nonhuman species and ecosystems in design processes.

Relationships between humans and nonhumans have been studied within design and computing as well. Most similar to the work described here is the concept of “animal personas” [33]. These animal personas have been proposed to account for species-specific considerations in the design of online systems used in animal agriculture, such as farming chickens. Similarly, “canine personas” have been put forth in the emerging area of Animal Computer Interaction, where technologies, such as digital emergency alarms, are designed specifically to be used by animals [86]. Raturi proposed the need for “system personas” [81]—a design concept that represents the system that the human is interacting with rather than the human themselves, and instantiated these as “farm personas” used in the design of digital technologies for sustainable agriculture. In the case of farm personas, the farms were detailed constructions of fictitious yet archetypal farms based on interviews with farmers and visits to small- to medium scale farms in California. Raturi demonstrated how such an approach can be used to develop system archetypes grounded in a critical analysis of real systems, and subsequently used in information design.

Beyond the connection to personas, other scholars have also engaged with nonhumans in design. DiSalvo and Lukens engaged with the role of nonhumans in design, providing an introduction to

concepts such as nonhuman expressivities, actor-network theory, and decentering [26]. Specific instances include Webber and her colleagues' description of how they engaged in design involving both orangutans and human stakeholders in the context of a zoo, to increase the well-being of the orangutans [99], as well as Laurier et al.'s exploration of the idea of dog-walking being a coordinated activity by both human and dog [54]. Kuznetsov et al. engaged with ethical issues surrounding DIY biology and "treating living organisms as inputs and outputs to digital technologies" [53]. Design ethnographer Anne Galloway has explored the interactions between media technologies and human/nonhuman relationships, offering a set of questions to provoke future work in this domain [37]. More broadly, the Animal-Computer Interaction conference [2] engages with a range of topics relevant to the ways humans and animals interact, in particular via computational technology. Our approach similarly engages with the relationship between humans and other living things, but with a particular focus on rectifying harm that has been done to nonhuman species in the past through human exploitation of ecosystem services.

Going deeper on the topic of decentering, various researchers have pursued efforts to decenter humans in different contexts. Smith et al. have sought to decenter the human in design processes [87], specifically seeking to understand the implications of increased interaction between humans and other animals (e.g., big cats). One instance they explore is how new designs, such as for "wildlife crossings"—safe paths by which nonhuman animals can cross human roads—could help make this cohabitation more viable. Similarly, Forlano has sought to decenter humans in the design of cities, in order to develop "more responsible, accountable, and ethical ways of engaging with emerging technologies" [31]. Our approach is aligned with these efforts, seeking to move humans out of a position of privilege.

Researchers in value sensitive design [13, 21, 35] have sought to understand processes by which values such as sustainability may affect design. Perhaps most related to the ecosystema concept explored here, Friedman and Hendry's "envisioning cards" [34] provide a mechanism for various stakeholders, values (such as environmental sustainability), and other factors to influence design processes.

Scholars within design and computing have sought to look beyond the human in various other areas as well. Akama et al. identified "plants, animals, rocks, rivers and spirits" as salient factors beyond the human in participatory design [3]. They provide thought experiments to help engage across the difference between humans and these "more-than-human entities". Coulton and Lindley's "more-than human-centered design" perspective [20] has explored similar perspectives as well. Lee's concept of the coevolution of humans and machines [55] offers the possibility that computational systems could be understood as new life forms; these systems could then represent another type of nonhuman for which to design. Liu et al. discussed working with nature, as opposed to controlling it, in sustainable agriculture, drawing on examples from how farmers engage with companion species [60]. They seek to broaden the design space in this domain by considering nonhuman stakeholders. Foth and Caldwell took a similar approach to media architecture [32]. Scholars have conducted workshops that engaged with more-than-human concerns in both human food practices [27] and the design

of smart cities [43], pushing back on the "human-exceptionalist" perspective embedded in much smart city research [43]. The work presented here shares a similar perspective with these works on thinking more broadly about the impacts of design beyond just the needs of humans.

Numerous HCI and design scholars have addressed environmental and sustainability issues in relation to topics such as interaction design [9], energy use [25], sustainable agriculture [80], and sustainable education [30], as well as cross-cutting analyses [6, 51, 67, 94]. Scholars have sought to push back against climate change and other environmental concerns via design framed around the Anthropocene, e.g., [56, 57], and have sought to envision a "good Anthropocene" [7] or to move toward post-Anthropocentric framings [103]. Nardi explored the role of design in seeking to address climate change [66]. Steven Umbrello's dissertation explored how nonhuman values could be integrated into design processes [95]. Thackara has proposed bioregioning as a model for enabling greater care for the biosphere [92]. These works align with the ideas presented here, similarly seeking to reverse the harms that human activities have caused. Nevertheless, many sustainable HCI efforts focus primarily on the benefits to humans of reversing these harms.

Despite previous efforts in related domains, the vast majority of work in computing and design fields ignores other species. By doing so, in the context of industrial civilization's default push toward the satisfaction of human needs and wants, this vast majority is implicated in systematically disenfranchising other species. Nearly all of the few articles that do engage with other species do so based on their relationship with humans (e.g., pets, zoos). The sustainability articles nevertheless frequently frame sustainability as being important because of its implications for humans.

### 3 APPROACH TO DESIGN KNOWLEDGE

Given that the vast majority of computing and design research fails to engage with the harm that humans are doing to the world's ecosystems, we discuss below the kinds of design knowledge, both in computing and beyond, that could contribute more effectively to addressing these issues. (We acknowledge that design knowledge is just one component of how particular designs may be created and implemented in the world—economic forces, individuals' motivations, and many other factors are relevant as well that lie beyond the scope of this analysis.)

We envision a future in which humans treat organisms from other species as having inherent value beyond their utility to humans (as well as having utility to humans via their role in the ecosystems on which human civilizations rely). This inherent value is a keystone of the deep ecology movement, mentioned earlier, which arose in the 1970s through the work of Norwegian philosopher Arne Næss [65] who was inspired by Rachel Carson's seminal work [16].

Humans have exploited other species for their own gain for millennia, and have done so with far greater technology-enabled power since the industrial revolution, and with even greater power since the rise of runaway wealth accumulation that accompanies neoliberal capitalism [41, 78].

*The future we envision is one in which this exploitation comes to be seen as problematic due to the inherent worth of nonhuman*

*organisms*. This vision is one in which humans have sole power among species to enact directed, intentional, system-level change, and are the only species that can accept responsibility for both the problems and potential ways to move forward.

In this vision, designers would reformulate their thinking around design, and engage with many different stakeholders, to consider human environmental impacts as a critical theme in *all design activities*. This reformulation would be driven by a variety of motivations, including justice for other species, as well as pragmatics for humans themselves.

Currently, designers, clients, users, and other stakeholders ignore many of the ecological costs to other species implicated in nearly all human design activities. By foregrounding these costs, and foregrounding the concept that most human flourishing comes at the direct and indirect expense of nonhuman species, future design activities can help enable system transformations that reverse human environmental impacts. Just as software designers have an obligation, at some point in the design process, to talk to clients about non-functional requirements such as safety and security (and now sustainability is beginning to be seen in a similar light [77]), so too should many different kinds of designers have an obligation to raise the issue of the ecological costs to other species of a particular design.

Non-functional requirements often create costs in the short-term (e.g., more programming time to build a secure password system). However, non-functional requirements reap long-term dividends (e.g., by preventing highly problematic security breaches). They provide value beyond the short-term costs, and often turn out to be absolutely essential to the success and sustainability of a system over time.

Although a post-growth economy—a “socially sustainable and equitable reduction of society’s throughput” [48]—has not yet been positioned as a non-functional requirement, doing so could be a crucial next step once sustainability is well-represented in design processes. Reversing human environmental impacts may create costs in the near future, but contribute to the goal of achieving longer-term and broader-scale societal benefits.

For example, imagine a designer eliciting requirements for a novel social networking system that requires continuous, high-definition streaming for all users. That designer could work with domain experts and many others to engage with the carbon emissions arising from such a system if it achieved market penetration similar to Facebook, and describe the implications of such emissions on climate change broadly as well as on particularly vulnerable species.

Note that, just as clients can overrule a software designer who suggests adding a password, clients can overrule designers with regard to environmental impacts as well. However, the requirement to raise the issue may still provide a useful contribution to enacting broad-scale system transformation, by confronting thousands of clients with this suite of issues and pervasively raising awareness of them. The human world is in a moment of global social ferment due to COVID-19 and other factors; while proposing such ideas might have seemed unrealistic even a very short time ago, humanity is now grappling with the agency of very powerful nonhuman forces (e.g., viruses) that scientists tell us are not going away.

### 3.1 Conceptual Design Tool: The *Ecosystema*

New kinds of design tools could support this vision. Here we describe a related conceptual tool that could provide a framework for the collection and sharing of design knowledge with the goal of reversing human environmental impacts. This theoretical construct is an adaptation of the idea of a persona: a description of a fictitious person used to support the development of products, marketing materials, etc. [69, 79]. We propose an additional archetypal entity: the *ecosystema*. An ecosystema is a tailored description of an ecosystem potentially impacted by particular design activities. This description should be adapted to the design process at hand, to provide actionable insights relevant to that process. We considered calling these “ecosystem personas”, but the term “persona” itself is typically anthropomorphic, so we opted for neologisms. While the personhood of nonhuman entities is part of various countries’ legal frameworks, as will be discussed later in this article, the common usage of the related term “person” typically refers to a human individual.

Just as a persona is not a comprehensive depiction of a particular person, but rather a conceptual lens through which to focus on particular aspects of a design space, an ecosystema is not a full biological description, but rather a construct highlighting aspects of ecosystems that are relevant to the area of work. Just as ecosystems may be spatially nested [50], various ecosystemas could also represent various spatial scale levels. Ecosystemas could be involved in the design process just as personas serve as tools for storytelling, for asking useful questions, for thinking about impacts, for creating empathy, and for keeping track of complexity.

### 3.2 Example: Technology for Regenerative Agriculture

Ecosystemas may be particularly relevant in the design of technology that influences both human-made and natural systems. Consider the growing domain of design research and technology development oriented toward supporting a transition to not only sustainable, but *regenerative* agriculture [5, 82]. The industrial agricultural revolution resulted in the intensive production of commodity crops and livestock, with ripple effects including decreasing water quality and soil health, loss of biodiversity and habitats, and impacts on climate change. Focus on short-term outcomes such as optimizing the upcoming year’s yield, and the growing practice of farming on leased land, has led to a complex set of issues in which human needs were prioritized over nonhuman species and ecosystems. While industrial capitalism tends to amplify and glorify the unidirectional removal of ecological wealth from ecosystems, alternative economic systems suggest that such large-scale exploitation is not inevitably linked to the provision of human material needs. On the other hand, farming practices, like those of the Amish community [8], for example, utilizes a set of practices that improve soil quality for agriculture over time. These practices have been in place for hundreds of years, and the Amish culture fosters their continuity across generations. In the past decade, there has been a push to adopt regenerative and sustainable agricultural practices such as reduced tillage, planting of cover and forage crops, integration of pollinator habitats, use of biological controls, etc., that aim to improve soil health, water quality, increase biodiversity, and sequester

carbon [91]. In short, the agricultural community is looking to provide a range of “ecosystem services” as shared threats to nonhuman species and ecosystems are intertwined with threats to agriculture and human food security. In this context, the provisioning of ecosystem services through agriculture may enable a reversal of past environmental harm that provides benefits to both human and nonhuman species alike.

The design of ecosystem incentive schemes and policies, practice verification systems, and digital tools to enable farmers to participate in digitally mediated payment exchanges, is a growing area research and development at the juncture of technology, policy, and regenerative agriculture [90]. Take the example of emerging digitally mediated Ecosystem Service Marketplaces (ESMs) that integrate lessons learned from the Payments for Ecosystem Services (PES) [52] and global carbon marketplaces [19]. The underlying idea behind ESMs is that they are digital platforms through which farmers submit farm data that demonstrate how their agricultural practices impact ecosystems and their inhabitants. ESMs ideally verify whether or not the agricultural practices truly render ecosystem services, quantify the level of service provided, and subsequently compensate successfully “regenerative” farmers, for instance, with monetary payments. Two startup companies, Nori [70] and Regen Network [83] are taking a blockchain-based approach for the verification of claims of ecosystem services provided through farm management practices. Nori, focuses on the certification of quantified soil carbon, offers a digital marketplace for the purchase of carbon offsets, and provides a cryptocurrency based on this carbon marketplace [71]. Regen Network, aims to verify landscape-level improvements (beginning with carbon sequestration) via remote ecological monitoring, and to provision a distributed ledger for decentralized data management, unified by a cryptocurrency [12]. Each company is engaged in a range of design work activities, both internally to improve their underlying blockchain technologies, as well as in collaboration with agricultural technologists and designers to support farmer data sharing via interoperability with farm management tools [102]. We envision that ecosystemas can provide valuable guidance to inform such design research and technology development.

In the design of an ESM, one may utilize a combination of traditional human personas, animal personas [33], and ecosystemas, as a means to capture the range of actors and systems that are influenced by the ultimate design of the ESM. If only personas were included, the ecosystem the farm purports to service would be voiceless. Thus, the *ecosystema* concept represents critical local ecosystems that are impacted by agricultural practices.

Consider a cattle ranch in the Midwestern United States that follows a deferred rotational grazing system, where herds of cattle are moved from pasture to pasture based on forage conditions. The ranch produces grass-fed beef for human consumption. The primary ecosystem services the ranch aims to provide are: to improve soil health and increase soil carbon, and improve habitats for local wildlife and native plants.

- *Personas*: Each of the major stakeholders of a ranch (e.g., rancher, farm workers, consumers, and buyers) are represented through personas. Though the primary agricultural decision maker, the rancher, determines what practices are

adopted on the ranch, farm workers are involved in data collection, and consumers and wholesale buyers make purchasing decisions based on agricultural data represented as certifications. Considerations (user requirements, desires, usability constraints etc.) can be made for each persona’s interaction with, for instance, a blockchain-based ESM. The different species that inhabit this space may similarly be given a voice through animal personas representing, for instance, birds that live in the riparian areas around the pasture, as well as the cattle themselves. Animal well being and welfare can thus inform design priorities in technologies for regenerative agriculture.

- *Ecosystemas*: Each of the adjacent and overlapping ecosystems that are directly impacted by the ranchers’ management decisions may be represented via an ecosystema. For instance, ecosystemas may be designed to represent: a perennial stream ecosystem (see Figure 2) (based on a real stream that runs through a particular pasture, but enriched with characteristics from similar streams in other regions), a coniferous forest (based on one that is adjacent to the ranch boundary), and the soil microbiome (based on one that underlays the entire ranch). Critical considerations, such as the verifiability of “improvements” to each of these systems via ecosystem services provided by the ranchers’ management decisions can now be considered. This may ground the design of a digitally mediated ESM that is technologically and scientifically feasible.

Ultimately, design constructs such as ecosystemas could help ensure that, for instance, agricultural technologies such as ESMs are designed with consideration for both human and nonhuman species in mind, and take into account the realities of both agricultural and ecological systems within spaces that we inhabit.

### 3.3 Creating Ecosystemas

Two specific artifacts illustrating the ecosystema concept (see Figures 1-2) were created by a team of faculty and students at a university. These students were drawn from both the fields of Computer Science and Ecology & Evolutionary Biology, and worked collaboratively on the design and content.

A factor the team explored in the design of ecosystemas was whether a particular ecosystema should be based on one real-world ecosystem (which we call a “simple” ecosystema) or on a hybrid of multiple ecosystems (which we call a “composite” ecosystema). A simple ecosystema would have all of its information drawn from a single real, spatially-delimited ecosystem. A composite ecosystema would be based on a single real ecosystem but potentially augmented by elements from other ecosystems (similar to how a persona may include aspects of multiple real people layered together into a composite representation). Both types would be tailored to the task of enabling human designers to consider the impact of their design on a range of ecosystems.

In this paper, all of the mock-ups are of simple ecosystemas. We determined that, at least at this early stage of thinking about ecosystemas, the biological correctness of a simple ecosystema was more important than the richness that a composite ecosystema would allow. However, it may be that, if/when ecosystemas become

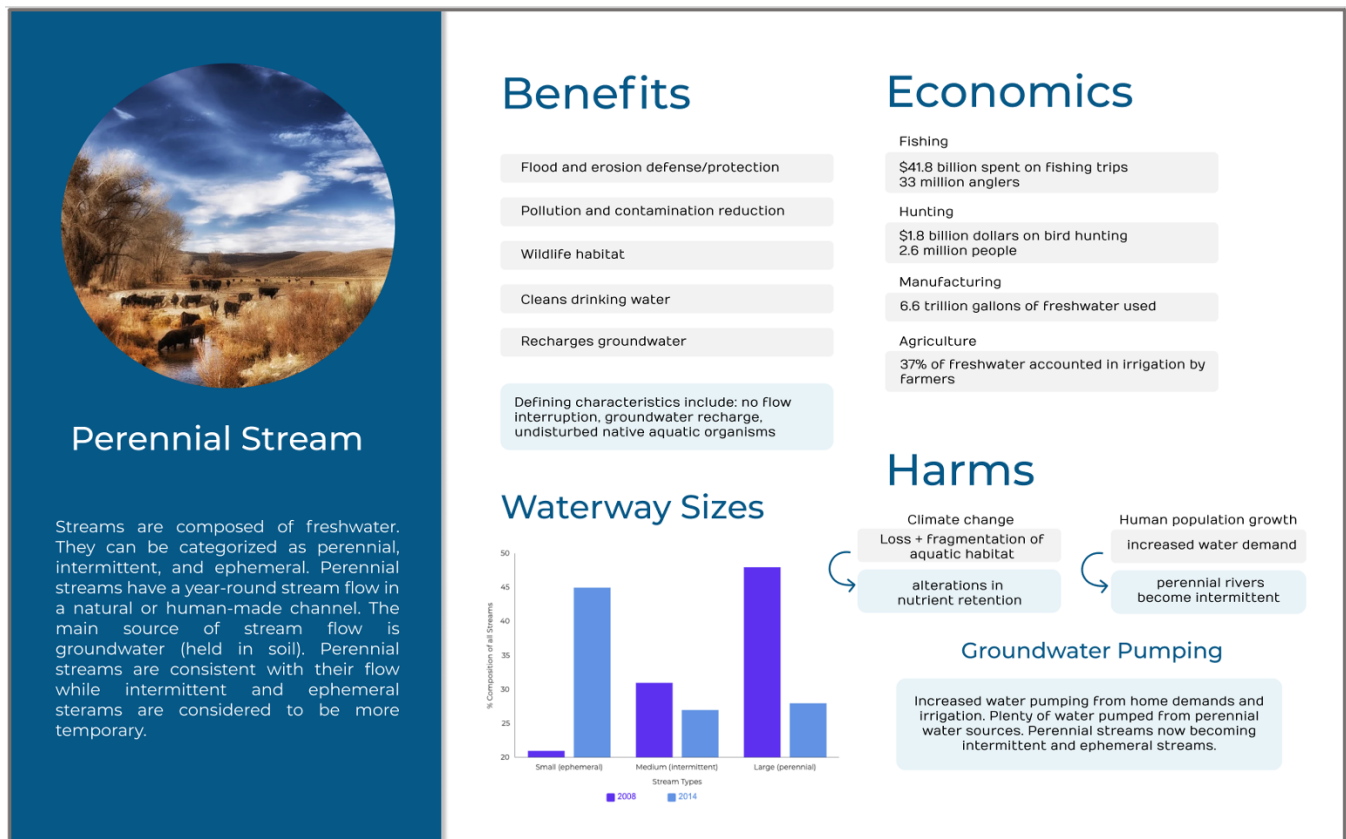


Figure 2: An ecosystema for a perennial stream.

more common in design processes, that both types may play a role in various design efforts. These examples are meant to be illustrative of two possible ecosystemas; in the text below, we present additional discussion of approaches to creating ecosystemas and deploying them in real worst contexts.

To ground ecosystemas in real world parameters, both should ideally be developed collaboratively by designers working with biologists or ecologists, and potentially many others with relevant expertise in engineering, business, etc. Designers may work with members of a population unfamiliar to them to develop a viable persona for the population. This process can help the designers keep the non-intuitive (for them) aspects of the population in mind while designing. Similarly, a designer could work with ecological experts on the constituents, interactions, and temporal patterns of particular ecosystemas to develop an ecosystema, as well as other experts regarding what aspects of an ecosystema would allow it to be more accepted and effectively deployed.

The concepts of ecosystemas point to the potential need for a new role, perhaps even a new profession—the *ecosystema advocate*. An advocate would be a person or group who has expertise about a particular ecosystema, and could be brought in as a consultant on design efforts. In current design activities a design team may engage with members of a target human population for various forms of consultation (focus groups, beta testing, etc.).

However, with nonhuman stakeholders, it may not be ethical or feasible to consult with particular nonhuman species or ecosystemas. With regard to ethics, there is often no clear mechanism for nonhuman species to be consulted by humans about what they would prefer, and it is likely many organisms could experience harm/fear if subjected to the impacts of alternative novel human designs (e.g., which forms of e-waste are least harmful to particular species [44], or which of two mobile phone tower deployments would be less disruptive to a particular habitat). With regard to feasibility, the logistical complexity of working with species that live under water, under ground, or in tree canopies could be prohibitive.

Therefore, a human who understands the purpose of design at a high level, and who can serve as an advocate for an ecosystema throughout the process, would be a valuable participant in design processes. Advocates have already been explored in other aspects of the design process (e.g., User Experience Advocates [42]). While many design efforts (e.g., the GWL urban eco-area project in Amsterdam [97]) already include environmental advisors, their engagement is often sporadic rather than central. Having ecosystema constructs that are familiar to designers could provide them with tools to think with in the absence of the consultants, and, in particular, when working with future projects in the early stages, before such consultants might be hired/involved.

In addition to collaboration among sustainability scientists/advocates and designers, it may be necessary to engage the participation of legal scholars as well so that protection of ecosystemas can be codified more concretely into formal law as a tangible, regulatory by-product of cross-disciplinary collaboration.

Designing to address environmental harm could be built on shared, evolving representations of ecosystemas, as well as other novel tools and techniques that can be used to integrate them with human needs and wants. These shared representations would require collaboration across fields as well as across industries. For example, the process of designing a food production system, which would need to engage with an ecosystema that reflects likely ecosystems where that food production system may be deployed, might produce design knowledge that would be relevant to an adjacent food distribution system, or to some non-food related system (e.g., a park, an energy infrastructure effort) that would engage with similar ecosystems. Clear, transferable documentation of ecosystemas will be valuable both in enabling different design activities, as well as bridging between design activities. This documentation process across multiple design scenarios could help derive and codify generic as well as situation-specific design guidelines to promote ecologically sustainable design. This generic vs. specific distinction may be linked to the notion of composite vs. simple ecosystemas; in the generic/composite case, there may be goals that are broadly shared across ecosystems (e.g., maximizing biodiversity), whereas in the specific/simple case, the survival of a particular endangered species may be most relevant.

Many different species and ecosystem concerns have to be weighed when trying to arrive at the most beneficial design strategies. Greater priority and advocacy may be warranted for species that are most endangered in a particular ecosystem and at a given time—but we also recognize the importance of sustaining non-endangered as well as endangered species, since the resilience of ecosystemas largely hinges on the stability of inter-relationships among the multiple, diverse organisms that comprise them [75]. Ecosystemas could be categorized along various dimensions (e.g., localized, regionally bounded, and global) and in terms of habitat location (e.g., fresh water, marine, terrestrial ecosystems). Ecologically-protective design decisions may need to be highly specific and adapted to particular regions, types of ecosystems, and the competing interests of the multiple species that comprise them. For example, within marine ecosystems, ocean carbonization and acidification are threatening the survival of a key “anchor species,” the single-celled phytoplankton, that, along with other marine photosynthesizers, produce one-half of the earth’s atmospheric oxygen through photosynthesis [63]. These species also serve as a major carbon sink when they die and sink to the ocean floor along with the carbon that they have absorbed. So, the “anchor” role of phytoplankton, given their foundational value for all other species residing in the oceans, may warrant assigning them greater priority relative to other species in their ecosystem in future design decisions.

Because of the value that these tools could provide across different design efforts, they would in some ways benefit from being similar to efforts at standardization. However, standards often do not adapt quickly to changing circumstances, and are in fact notoriously slow-moving. For example, standardization processes undertaken by the International Organization for Standardization typically take

several years to complete [47]. While informal adoption of models of ecosystemas could happen more quickly, the process of enabling revisions to those models to propagate across different organizations would be important as well. Given the need for rapid response to the environmental crises that currently face humanity and the ecosystems we inhabit, enabling ecosystemas to be updated rapidly would be very beneficial.

Another approach could involve the development of patterns for ecosystemas’ use [4]. There may be various contexts in which ecosystemas are brought to bear differently; patterns could help provide guidance to teams interested in working with ecosystemas. Just as Alexander [4] has created a “pattern language” of design guidelines intended to improve the quality of human-made environments, it may be possible to codify an analogous set of ecosystema patterns—a set of generic best practices for optimizing the various facets of ecosystemas and their component species.

Given the complexity of ecosystem-level design, and therefore the near-certainty that the design team’s understanding of the system would in some way be flawed, it would also be relevant to design in a way that allows for dynamic revision based on the evolution of the ecosystemas on which the design was based. What would unfold would be a process of continuous accountability [28], continuous improvement [93], and an iterative learning process/complex adaptive learning system similar to those currently found in healthcare [36, 74]. As new versions of software, hardware, and other products and service were being developed, salient revisions to ecosystemas that had been associated with earlier versions would be flagged for designers to consider in the redesign process.

Nevertheless, for those common goals to include the good of nonhuman species and ecosystems in a nontrivial way, there would need to be some substantial shifts in priorities for most corporations. Current models of capitalism are largely structured around benefits that accrue to shareholders (i.e., shareholder primacy [84]). However, voluntary frameworks such as corporate social responsibility [18] and triple bottom line accounting [29], provide some hope that capitalism may be able to shift its focus. In addition, recent proposals around post-growth and degrowth [48] provide intellectual framings for such a transition.

### 3.4 Representation for Nonhumans

In order for the needs of nonhuman species to be well-represented, those species need to be given leverage of some kind. For example, there could be financial accounts managed by their advocates, or distributed autonomous organizations [72] that algorithmically serve their needs. Or there could be legislation (for example, expanding on existing legislation such as the US Endangered Species Act) that provides other species and ecosystems with non-trivial power of some form, to create systems that benefit them, to block efforts that would disadvantage them, or both. Whatever form it might take, this leverage may impact many different aspects of civilization, including design.

Giving other species and ecosystems concrete leverage is largely at odds with current norms in the industrial economy. In addition, there are deep philosophical questions that arise about the relationship between leverage for nonhumans and questions of nonhuman cognition [76]. Nevertheless, giving various species and

ecosystems agency in legislative, economic, and design processes is not unprecedented in human civilizations. Nonhuman-animal personhood is being explored in the US and around the world [49], as are questions of whether nonhumans have legal standing [1] and the right not to suffer [38]. In the domain of ecosystems, the Whanganui River has legal personhood in Aotearoa/New Zealand [68]. Legally, the river has various rights, and is represented in negotiations by two human guardians. Similar laws exist in several other countries, including Ecuador, Bolivia, India, and the United States [39]. Corporations in the US have legal personhood which gives them substantial leverage on governmental processes; extending personhood to nonhuman species and entire ecosystems could provide those entities with similar leverage over the human systems that are so profoundly impacting them.

## 4 FUTURE WORK

While the ecosystema constructs could be a useful part of ecologically sustainable design processes that seek to take into account the short- and longer-term needs of multiple species and the ecosystems they reside in, future work to understand the specific content of such design guidelines would be very valuable. Some aspects of such standards could be generic, that is, applicable across a wide range of ecosystema design scenarios. Other aspects may be more context-specific, addressing the particular circumstances posed by each ecosystema design situation (e.g., the array of ecosystem services challenges present in that particular setting).

As the world's circumstance becomes worse environmentally, the default approach for much of humanity will likely be to more rapidly devastate ecosystems in the relentless quest for resources. As such, the human predicament may be intractable. Nevertheless, there are moves afoot such as degrowth [48] that, while not without their challenges, suggest potential directions for realigning human systems. Reversing anthropogenic environmental harm is just one idea for one aspect of this very complicated problem.

Looking more broadly still, it may be that the collapse of human systems would be even more beneficial for other species than any effort by humans to reverse anthropogenic environmental harm. To return to an example mentioned earlier, while the Chernobyl nuclear disaster made that region unusable for humans, it has not been an equivalent disaster for nonhuman species. In an article titled "Long-term census data reveal abundant wildlife populations at Chernobyl", a team of researchers writes: "our long-term empirical data showed no evidence of a negative influence of radiation on mammal abundance." [22] Similarly, major human failures such as failed states are likely only failures from a human perspective. A sweeping failure of human civilizations could set the stage for nonhuman ecological recovery [100]. An exception would be if the civilization failed because it destroyed its own ecosystem, in which case it may not be possible for the ecosystem to recover. For example, after humans cut down all the trees on Rapa Nui (also known as Easter Island), the island was bare of trees for hundreds of years (until recent reforestation efforts) [62].

Therefore, even the act of attempting to conceive of design knowledge to reverse human-caused environmental harm (rather than simply allowing human civilizations to collapse as soon as possible) may be vexed. However, being constrained by our own

existences, which love and value family and friends, and which see humans as a species equally deserving of life and well-being as other species, we nevertheless advocate for efforts to engage in system transformation through design, rather than simply accepting that the collapse of human civilizations may be the most "desirable" alternative from a perspective that does not prioritize the experiences of humans.

## 5 CONCLUSIONS

In this article, we have presented a potential approach to design and design knowledge that may help to reduce and reverse anthropogenic environmental harm, in particular through the use of ecosystemas. We hope that someday, a large enough portion of humanity will see nonhuman species as being of sufficient inherent value to enable the deep cultural, economic, and political shifts to bring about futures that are beneficial not just for humanity, but for all other forms of life. Caring for others is seen as a great good by nearly all humans and all human communities. For some people, this only means caring for their family, or all people "like them" in whatever way they define "likeness". For others, it means caring for all humans. For a few, it means caring for all living things. But the fact that caring for others is a core aspiration for most people gives hope that perhaps the scale at which that caring is enacted could be shifted [101]. And as such a shift comes about, we hope that designers will use their great enthusiasm and prodigious talents to transform systems and bring forth viable alternative ways of being in the world.

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